



Geographical Concept

GEOG 1026

Department of Geography
National Taiwan University

The Geographer, 1668-1669, Johannes Vermeer

Course Objectives

- Maps are the language of geography. Cartography is the science, technique and art of creating maps. It involves collecting geographic information, selecting and simplifying data, and producing maps that communicate spatial patterns, relationships, and geographic phenomena.
- Geographic information systems (GIS) are computer-based systems to collect, model, process, analyze, and visualize geographic information.
- This course integrates theory and practice by introducing the principles of cartography and the theoretical foundations of GIS, strengthening spatial thinking, and cultivating students' ability to explore geographic issues using geographic information.

What is geography?

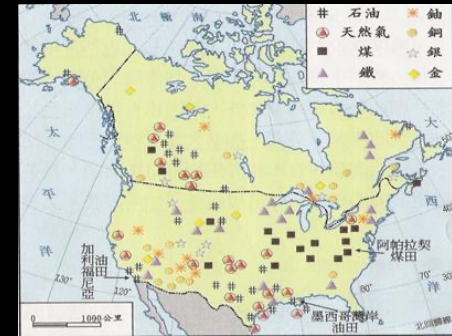
The Geographer



What society thinks I do



What academy thinks I do



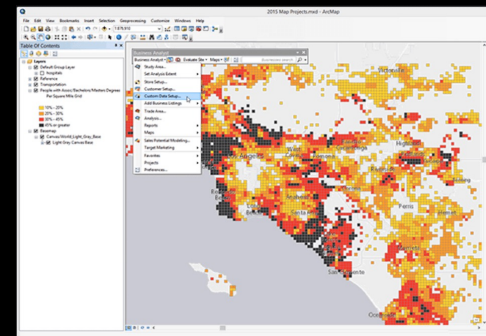
What my mom thinks I do



What friends think I do



What I think I'm doing



What I actually do



The Geographer

Johannes Vermeer
1668-1669

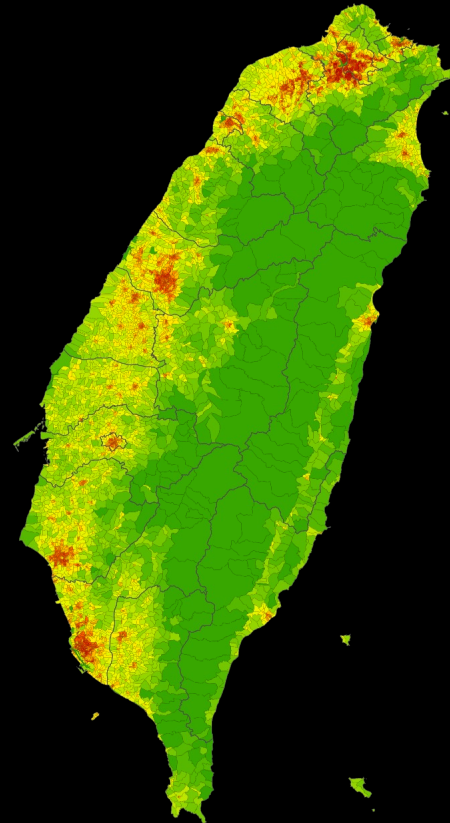
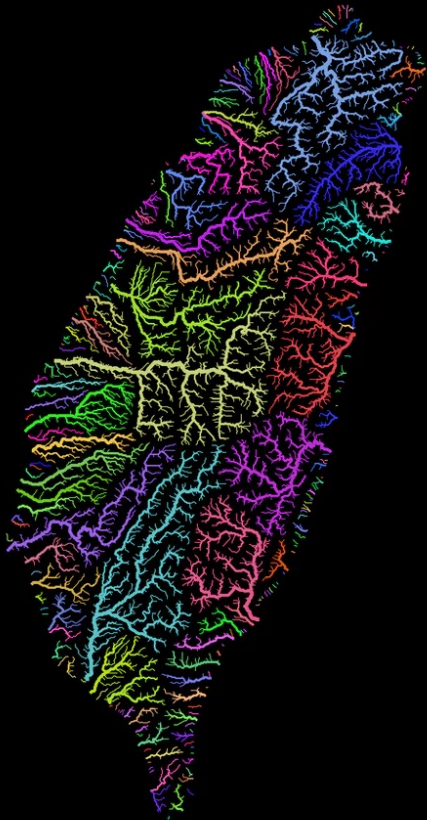
"someone excited by intellectual inquiry"

Geography

- Geo + graph
- The study of the Earth as the world of humankind, with particular emphasis on **spatial** phenomena
- Four traditions of geography (Pattison, 1963)
 - Earth science tradition
 - Spatial tradition
 - Area studies tradition
 - Man-land interaction tradition

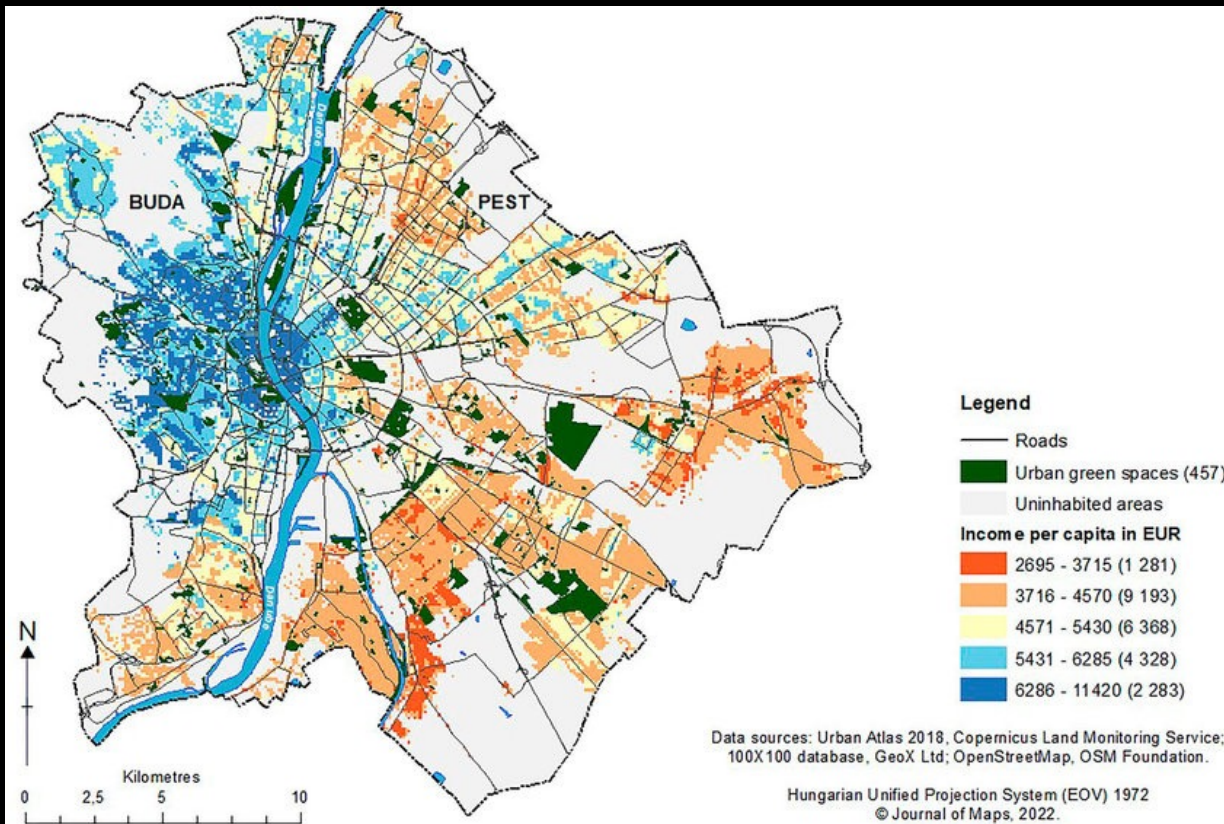
Geography

- Physical geography vs. human geography?

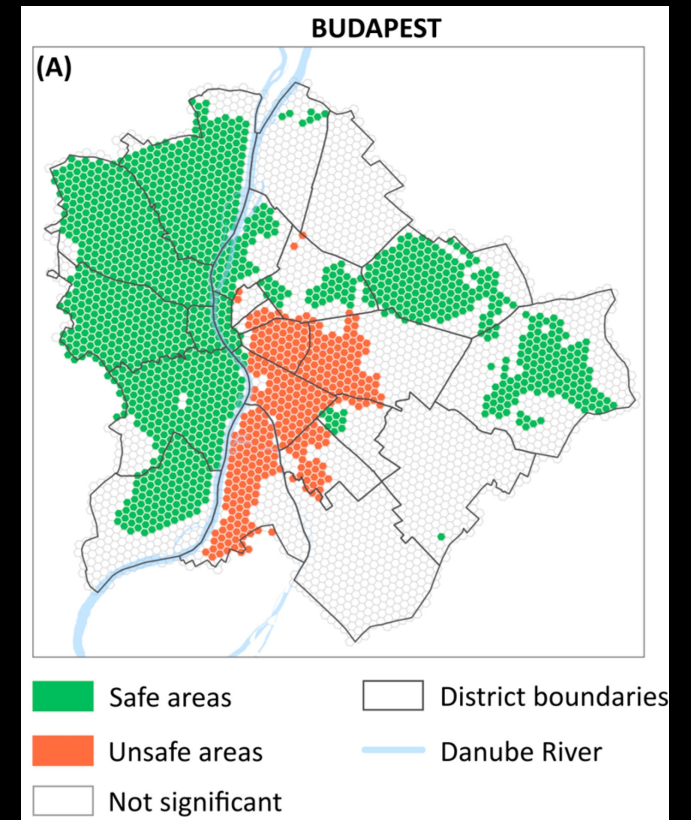


Geography

- Quantitative vs. qualitative?



Farkas et al. (2022)



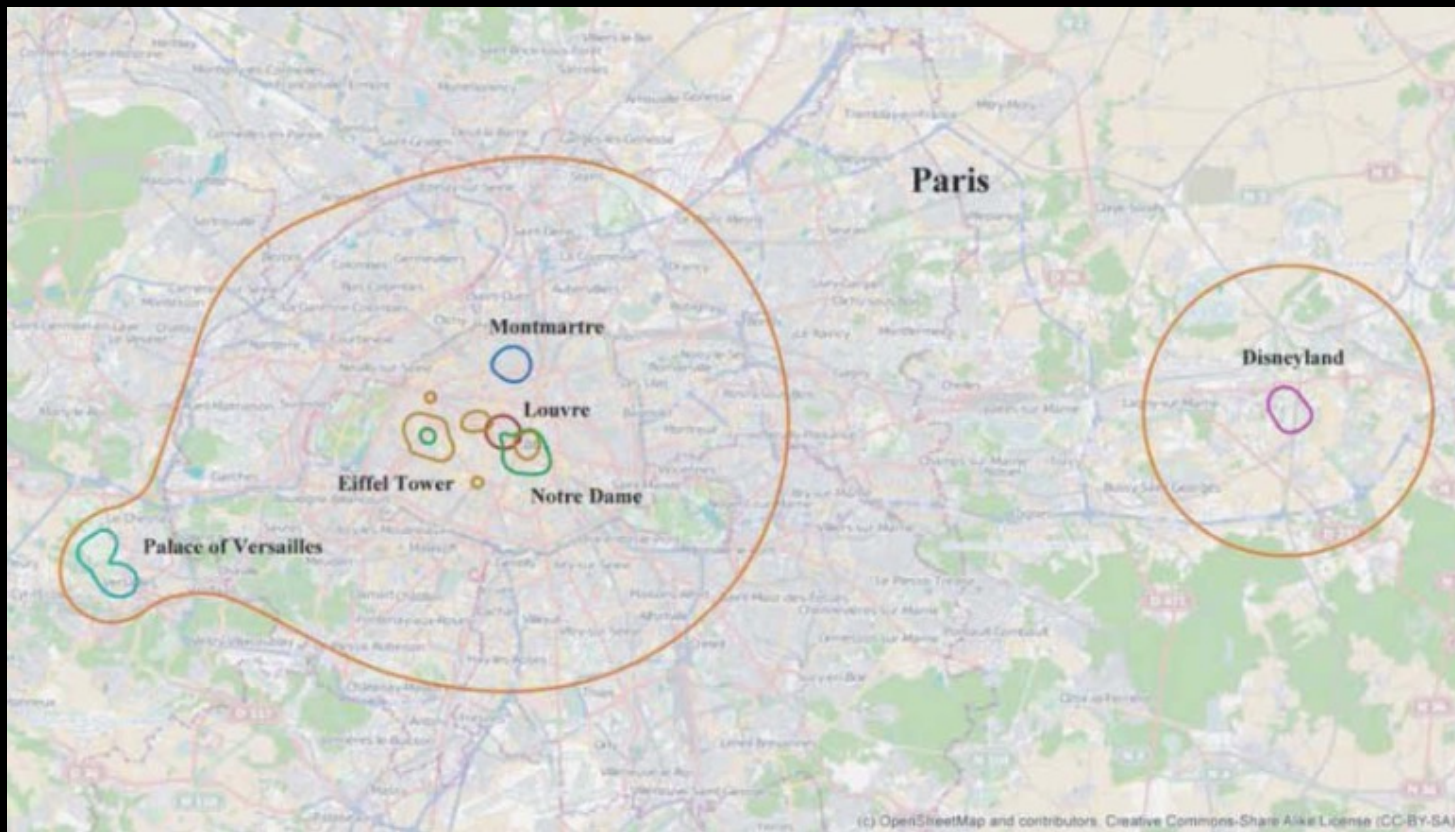
Kounadi et al. (2025)

Geographical Concepts

- Space
- Place
- Scale
- Change

Geographical Concepts

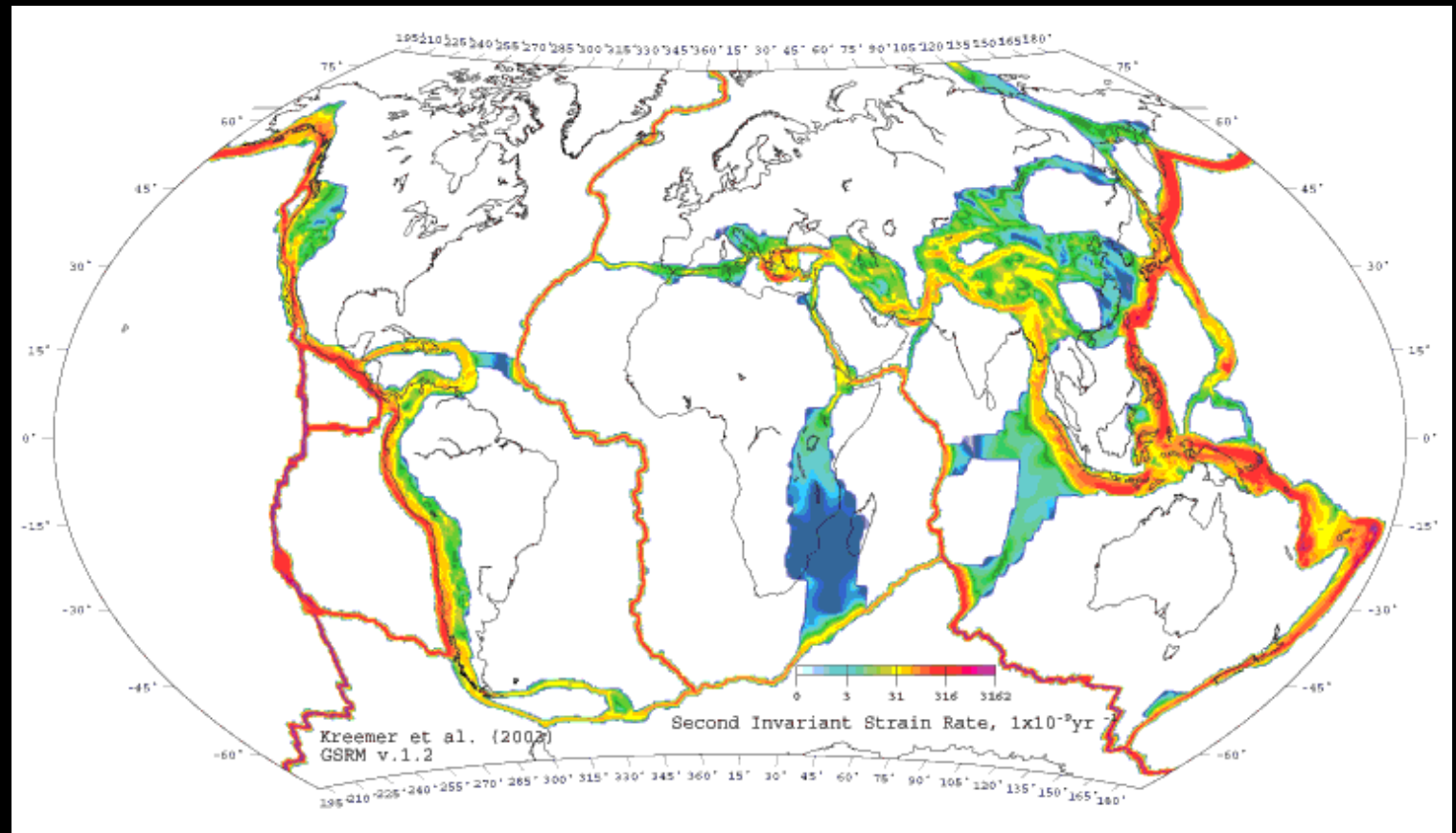
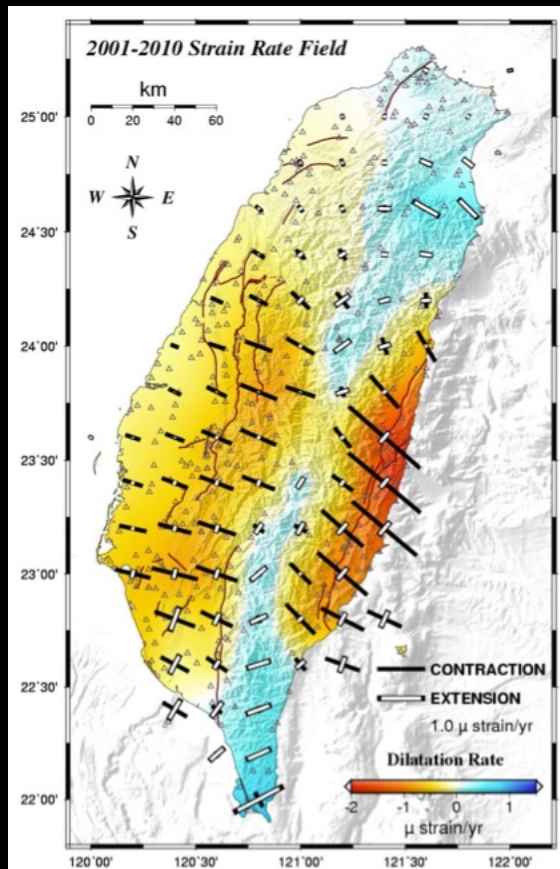
- Space vs. place?



Longley et al. (2014)

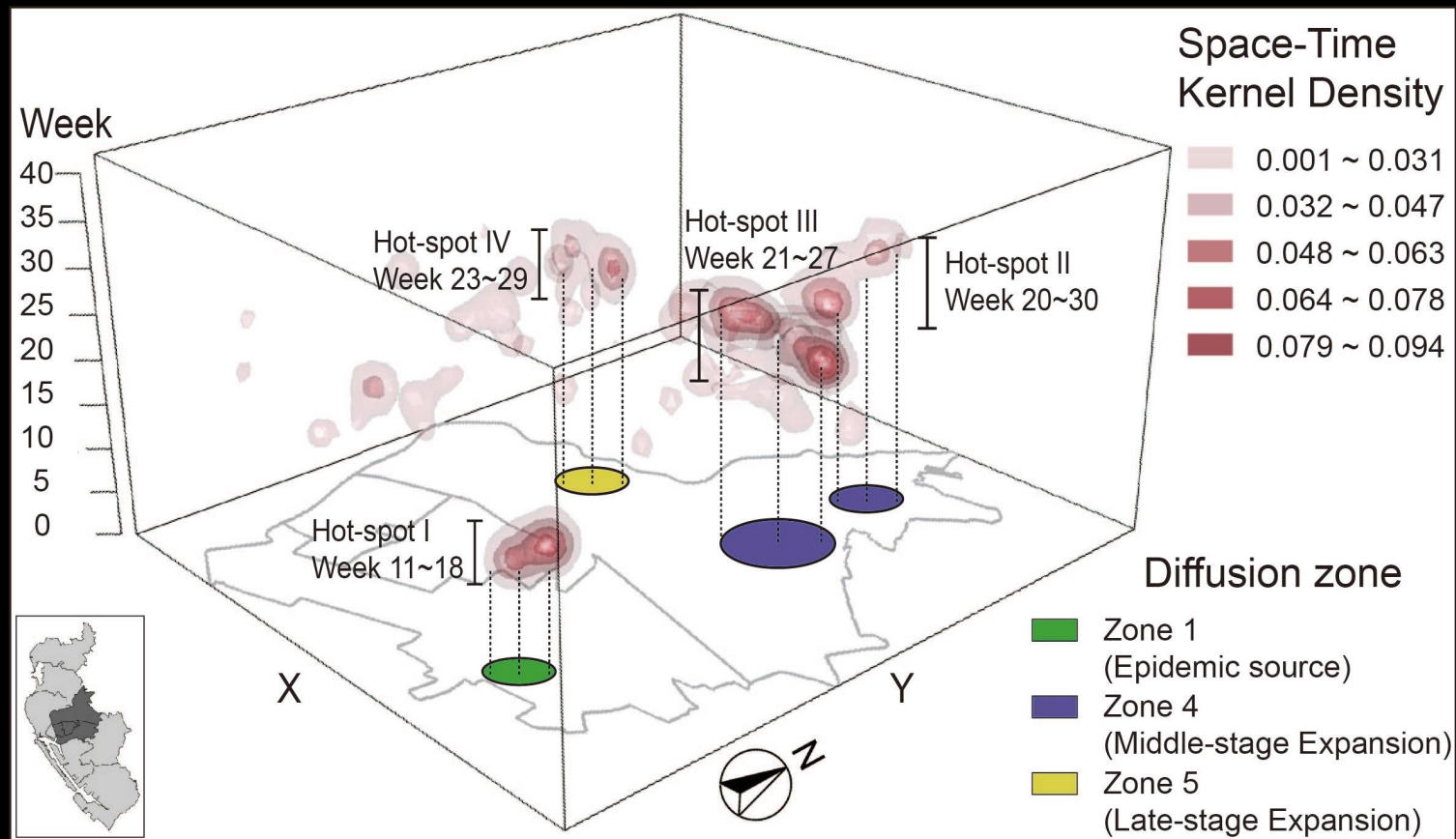
Geographical Concepts

- Local vs. global?



Geographical Concepts

- Spatiotemporal change

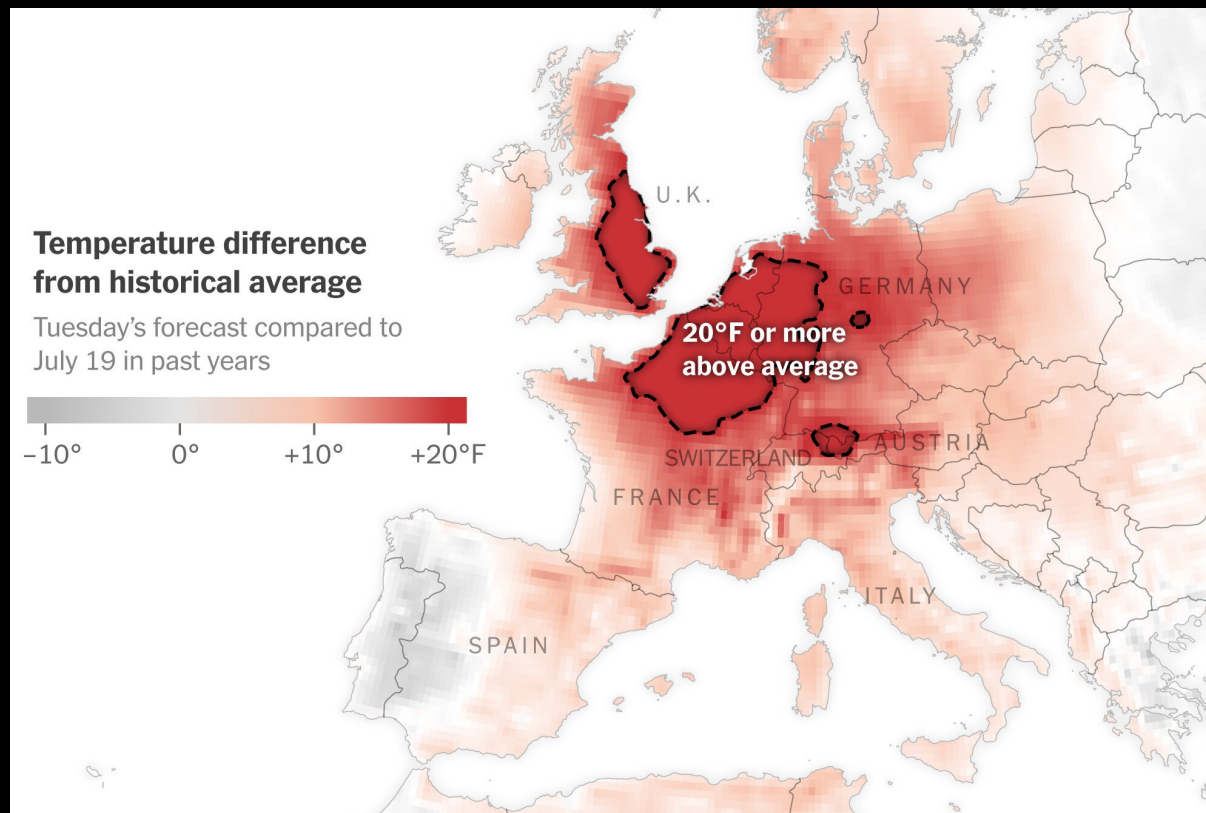


Laws of Geography

- Tobler's first law of geography:
 - Everything is related to everything else, but near things are more related than distant.
- Tobler's second law of geography:
 - The phenomenon external to a geographic area of interest affects what goes inside.
- Arbia law of geography:
 - Everything is related to everything else, but things observed at a coarse spatial solution are more related than things observed at a finer resolution.

Laws of Geography

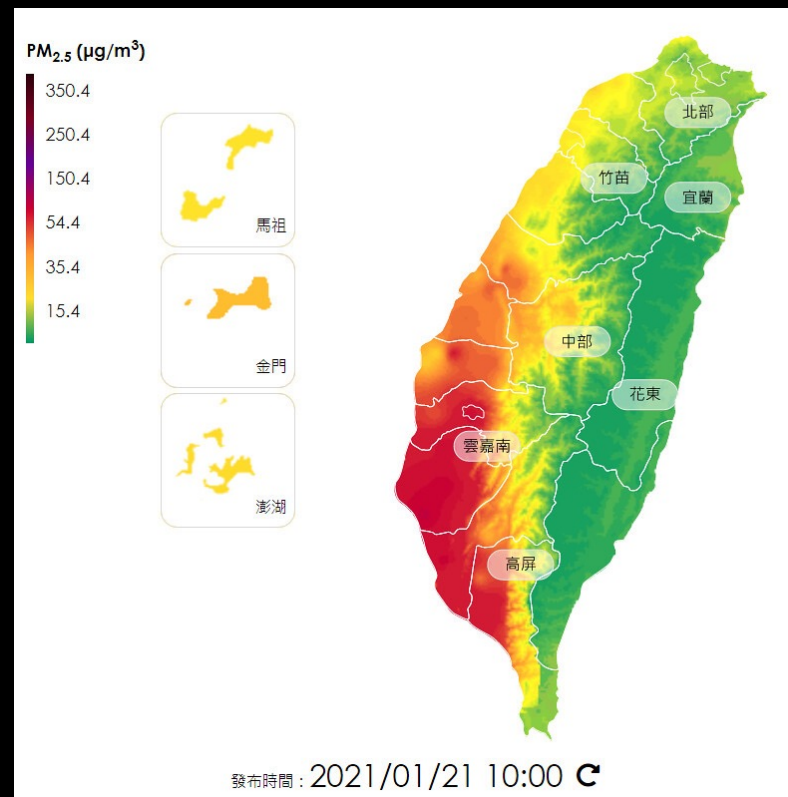
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<https://www.nytimes.com/interactive/2022/07/19/world/europe/uk-europe-heat-map.html>

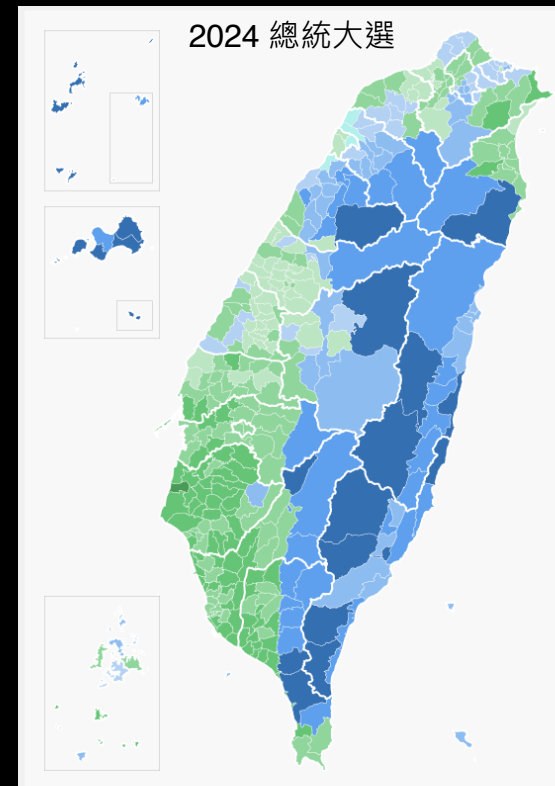
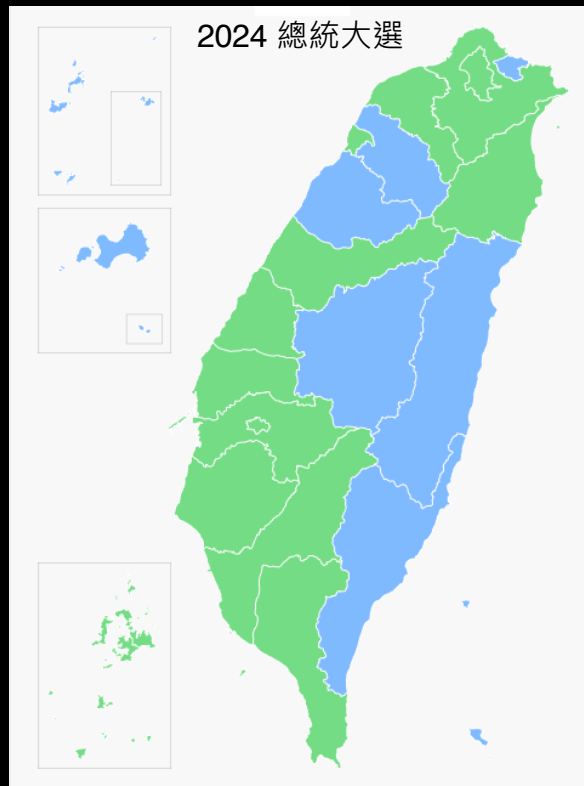
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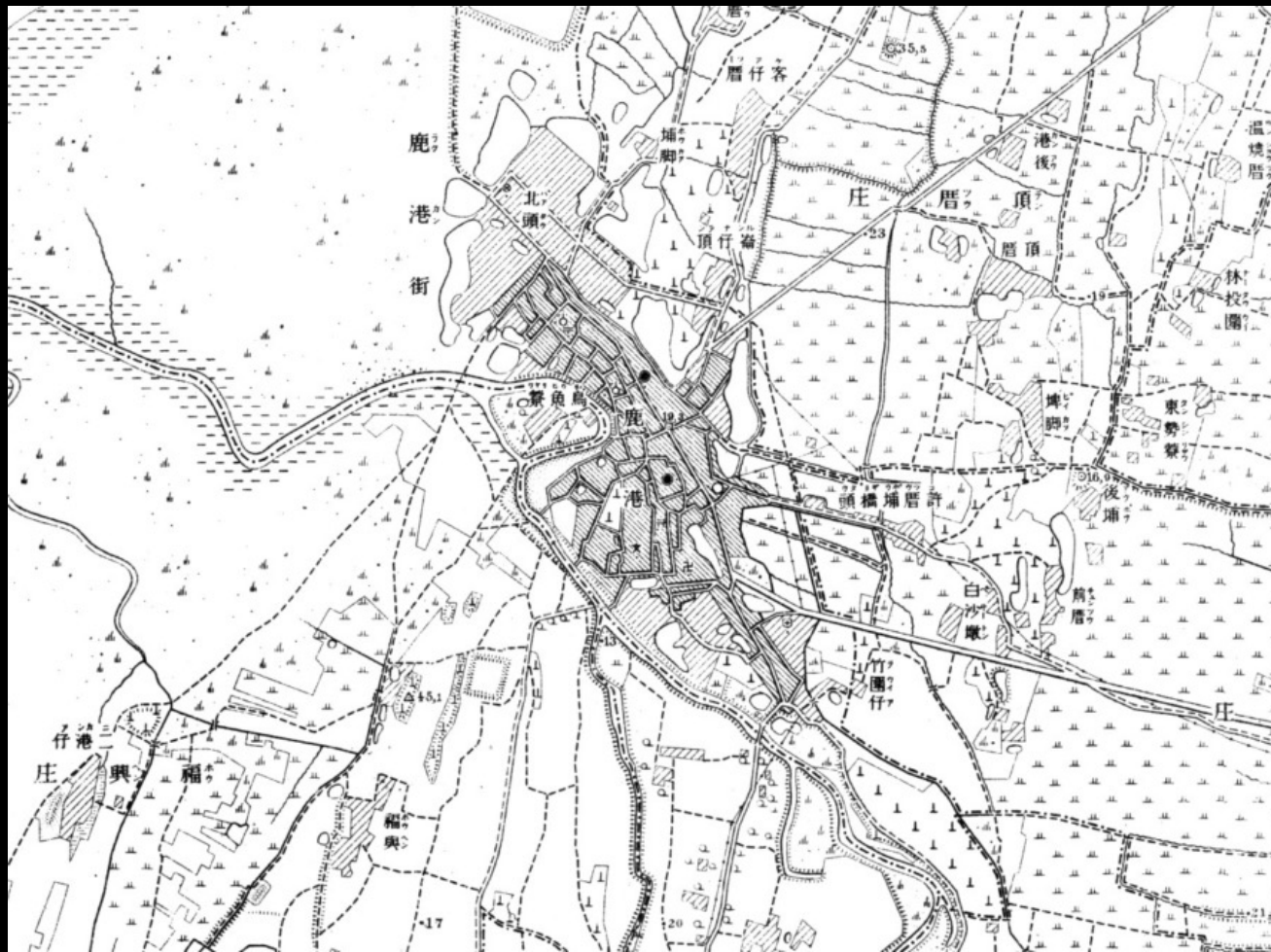
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Laws of Geography

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日本物理學家、散文家寺田寅彦
(1878-1935) :

現在，假設取地形圖中的任意一角落，而必須將其中滿載的各種知識，翻譯成我們的「日語」，那將是件大工程。光是一條等高線的曲折，就幾乎不可能以一支筆道盡。但是用「地圖的語言」，只要一眼就能把土地的高低起伏、坡面的緩急等形象印在腦海中。不只如此，甚至能獲得大小道路的連絡、山坡樹林的樣貌、耕地的分布與種類的概念。

MIT Sloan

Management Review

(2012)

Interview with Simon Thompson (Esri)

Location Analytics: Bringing Geography Back

≡ MENU
**Harvard
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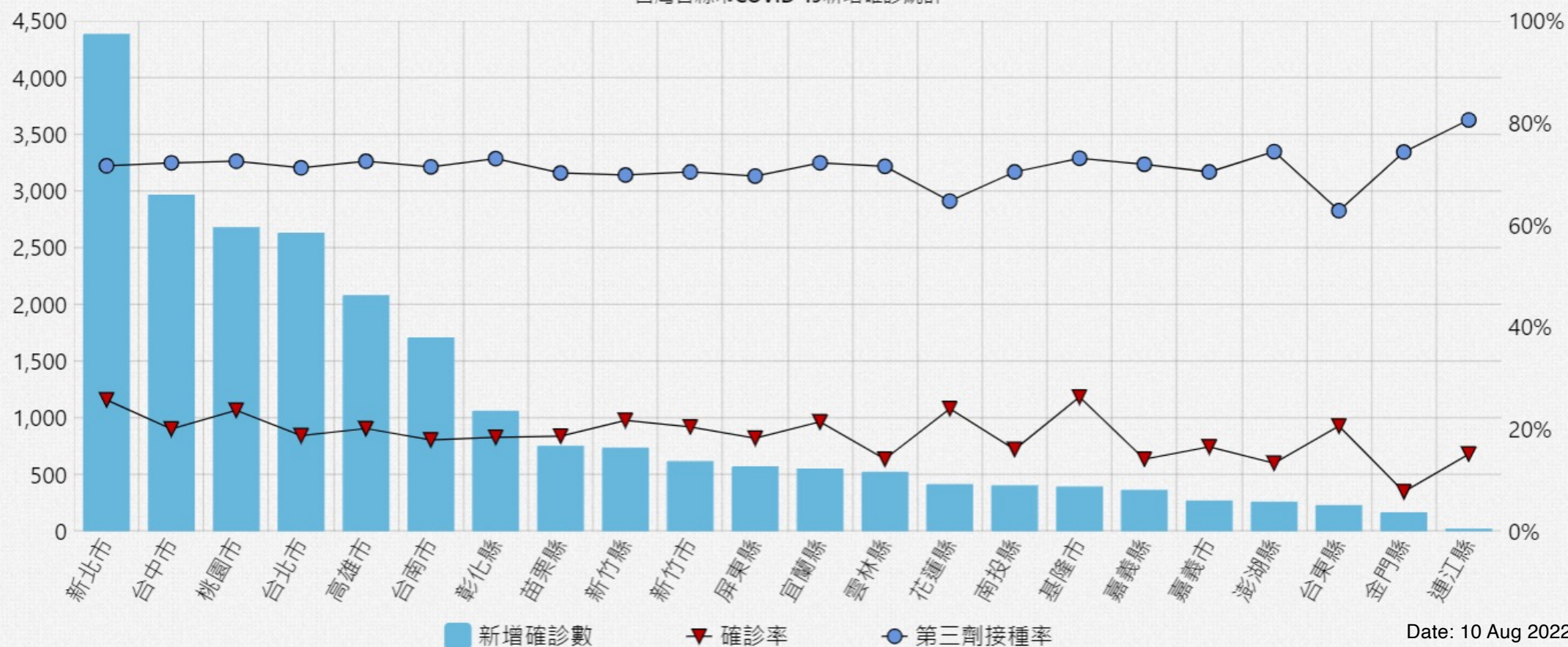
MARCH 12, 2014

**What counties have higher COVID-19 cases
on 10 August 2022?**

**Solution: Listing COVID-19 cases in a table
or chart**

各縣市區域每日新增確診圖表

台灣各縣市COVID-19新增確診統計



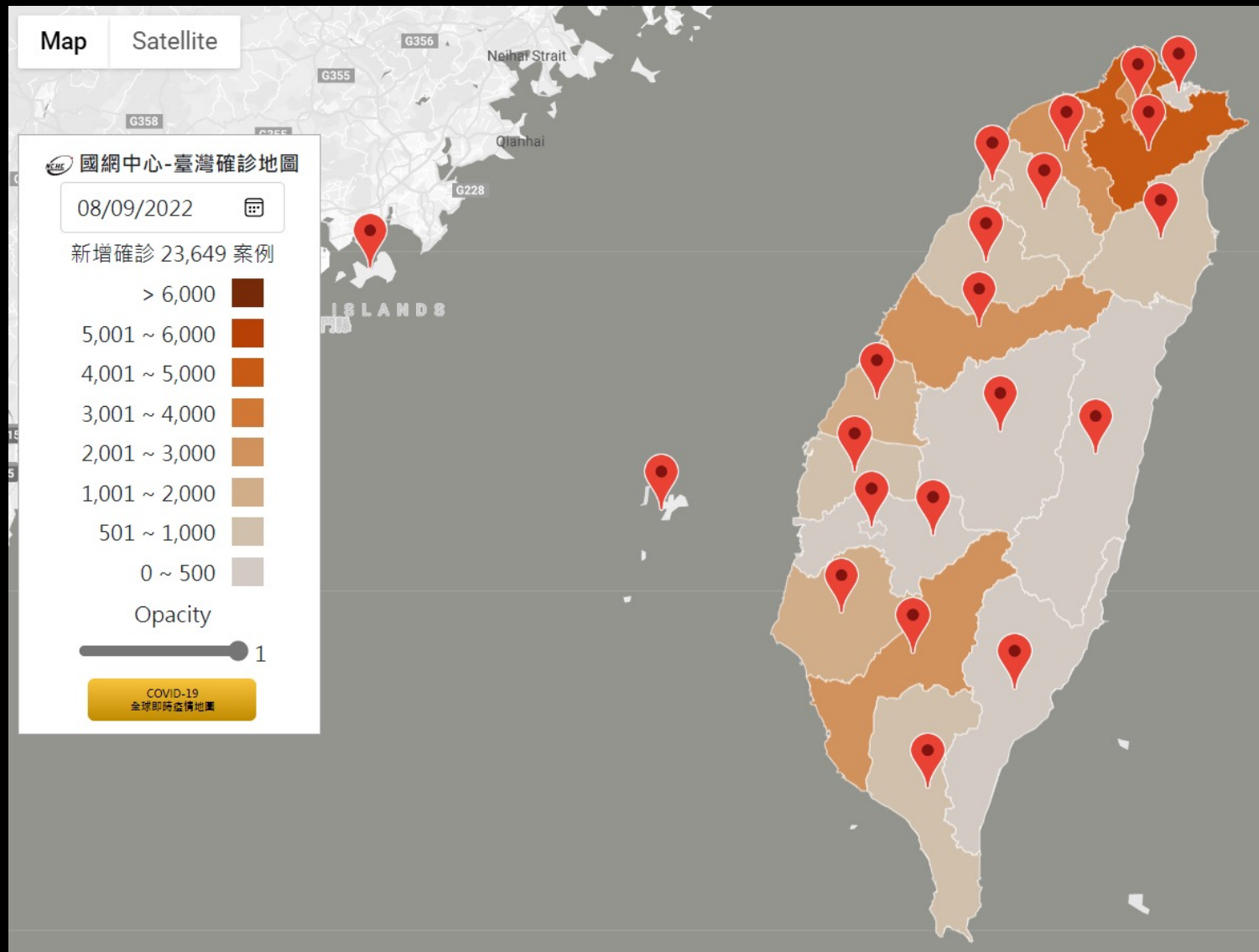
Date: 10 Aug 2022

https://covid-19.nchc.org.tw/city_confirmed.php

What counties have higher COVID-19 cases on 10 August 2022?

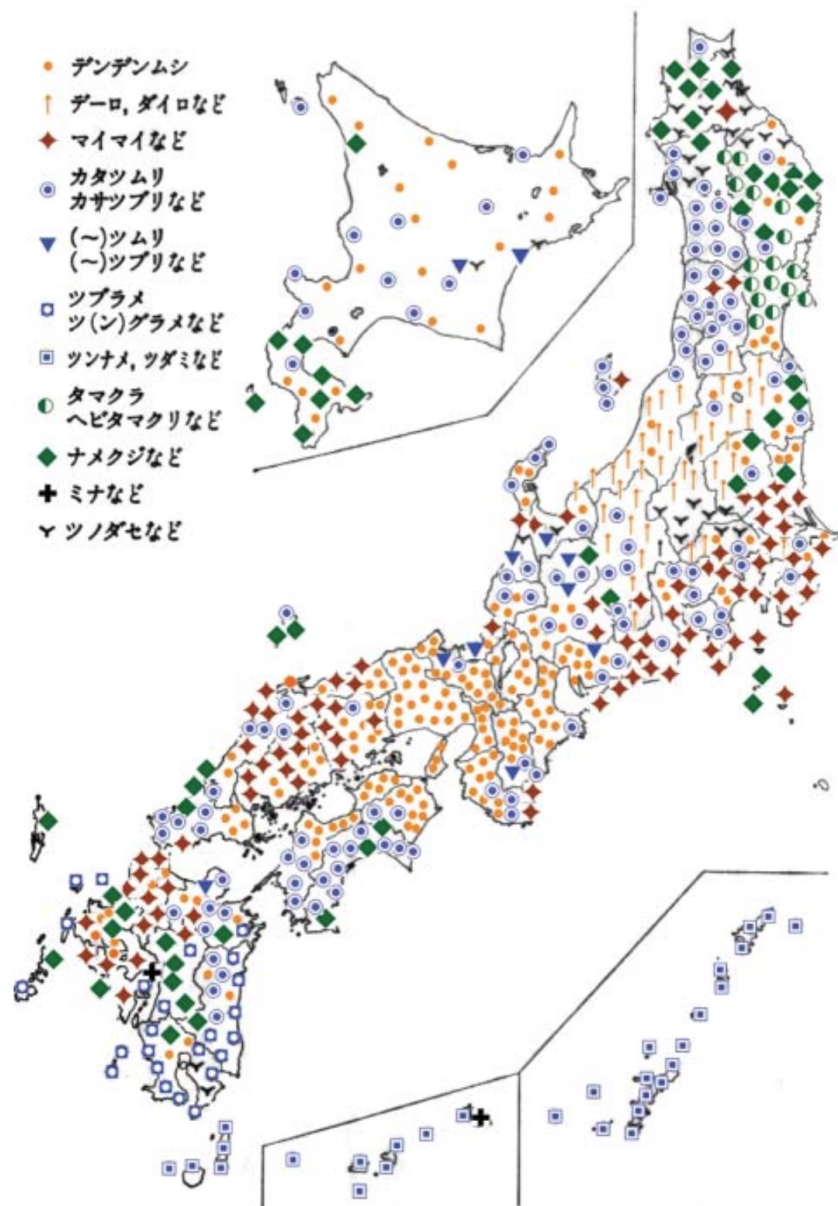
Solution: Listing COVID-19 cases in a table or chart

Solution: Showing COVID-19 cases in a map



https://covid19.biobank.org.tw/narl/city_today.aspx

Let's see more examples!



How do dialects evolve in different regions?

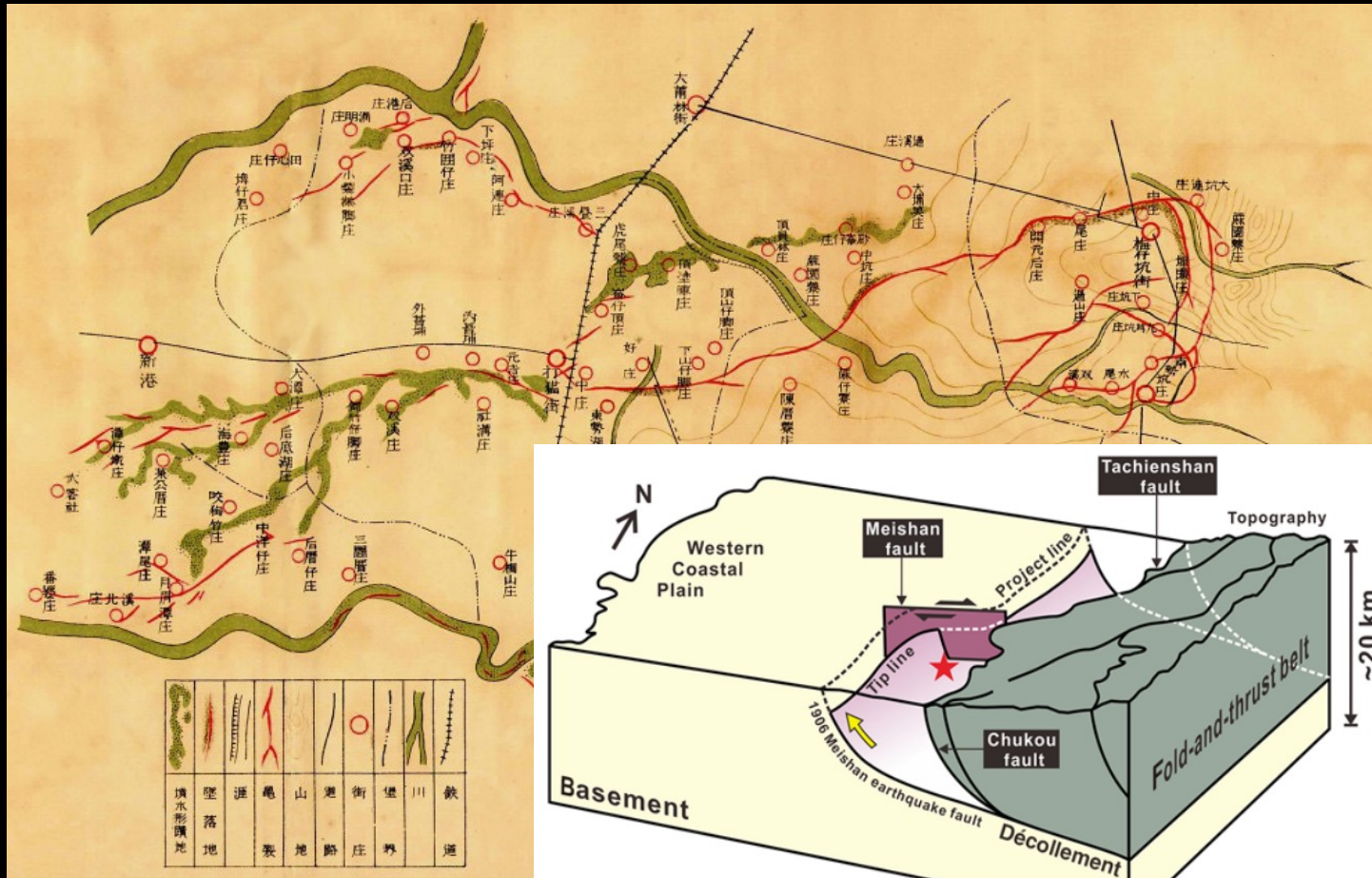
蝸牛方言地圖

編號	柳田分類	徳川分類	顏色
E	デデムシ系	デンデンムシ (dendenmushi)、デーロ (de-ro)、ダイロ (dairo)	橘色
D	マイマイ系	マイマイ (maimai)	褐色
C	カタツブリ系	カタツムリ (katatsumuri)・カサツブリ (kasatsburi)	藍色
B	ツブリ系	(〜)ツムリ (~tsumuri)、(〜)ツブリ (tsuburi)、ツブラメ (tsuburame)、ツングラメ (tsungurame)、ツダミ (tsurami)	
A	ナメクジ系	タマクラ (tamakura)、ヘビタマクリ (hebitamakuri)	綠色
其他	ミナ	ミナ (mina)、ツノダセ (tsunodase)	黑色

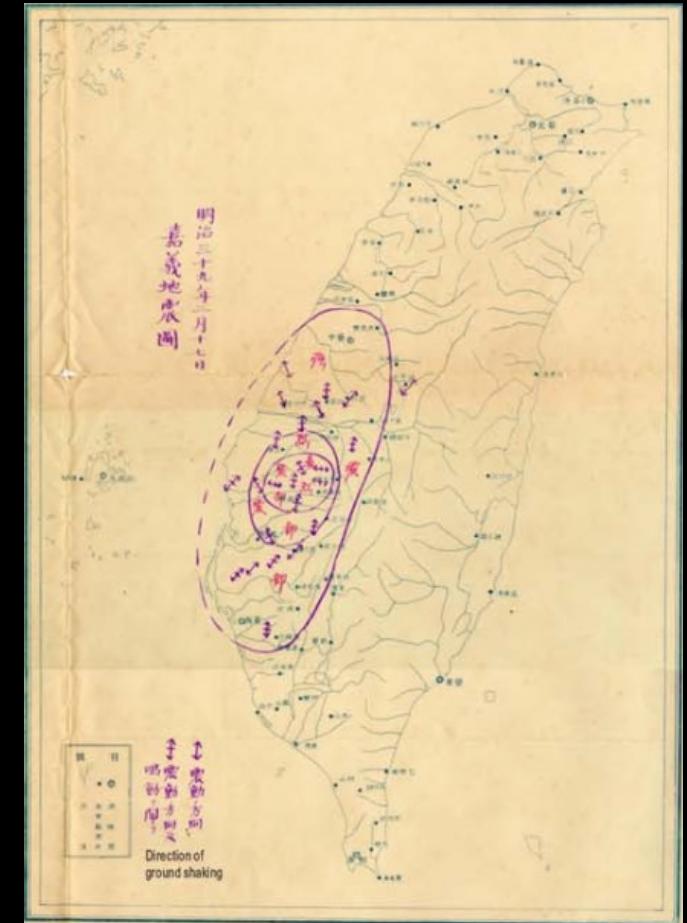
洪惟仁 (2015)・改繪自柳田國男(1930)與徳川宗賢(1974)

柳田國男考證各種變體的歷史，發現最古老的變體是 A，然後 B→C→D→E，E 是近畿最近產生的新變體，所有變體都是由京都往外層層擴散，越遠離近畿，保存著越古老的變體。柳田國男因此提出有名的「方言周圈論」，主張「古語殘存於邊境」，也就是說方言地理變異，周邊一致性的變體是存古的，中間的變體是後來插入的新形式。

What is the source of the 1906 Meishan earthquake?

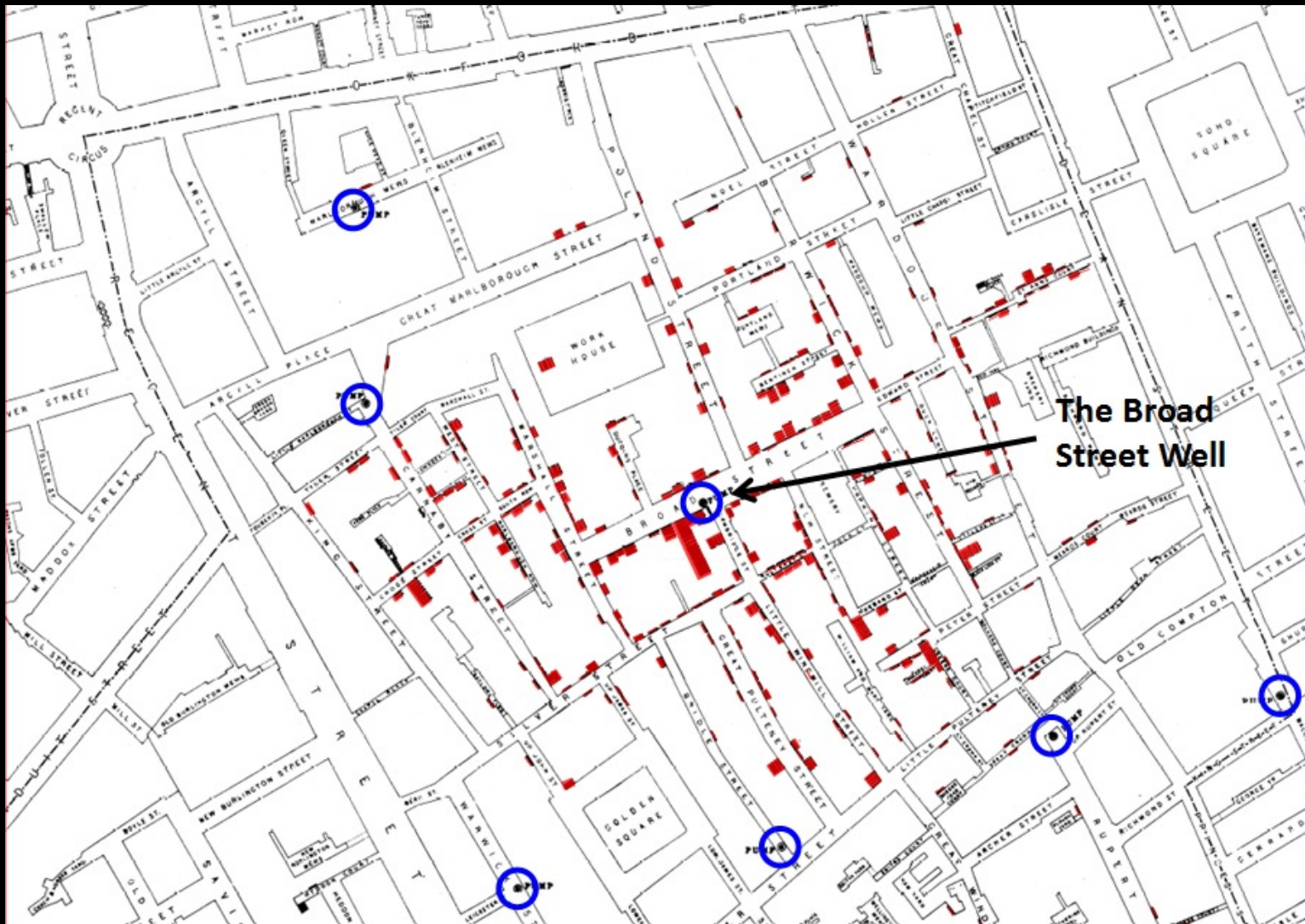


▲ **Figure 9.** The scheme for the geometry of the possible episode for the E-W-striking surface rupture between two N-S thrust faults. The star represents the hypocenter of the 1906 Meishan earthquake. The color version of this figure is available only in the electronic edition. (Liao et al., 2018)

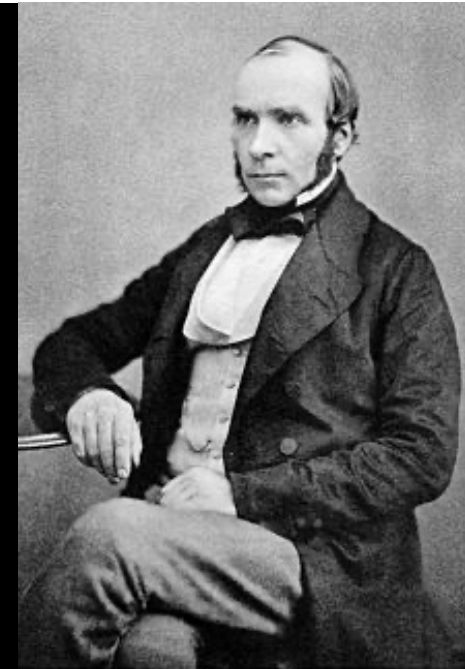


(台灣總督府, 1906)

How does cholera spread?



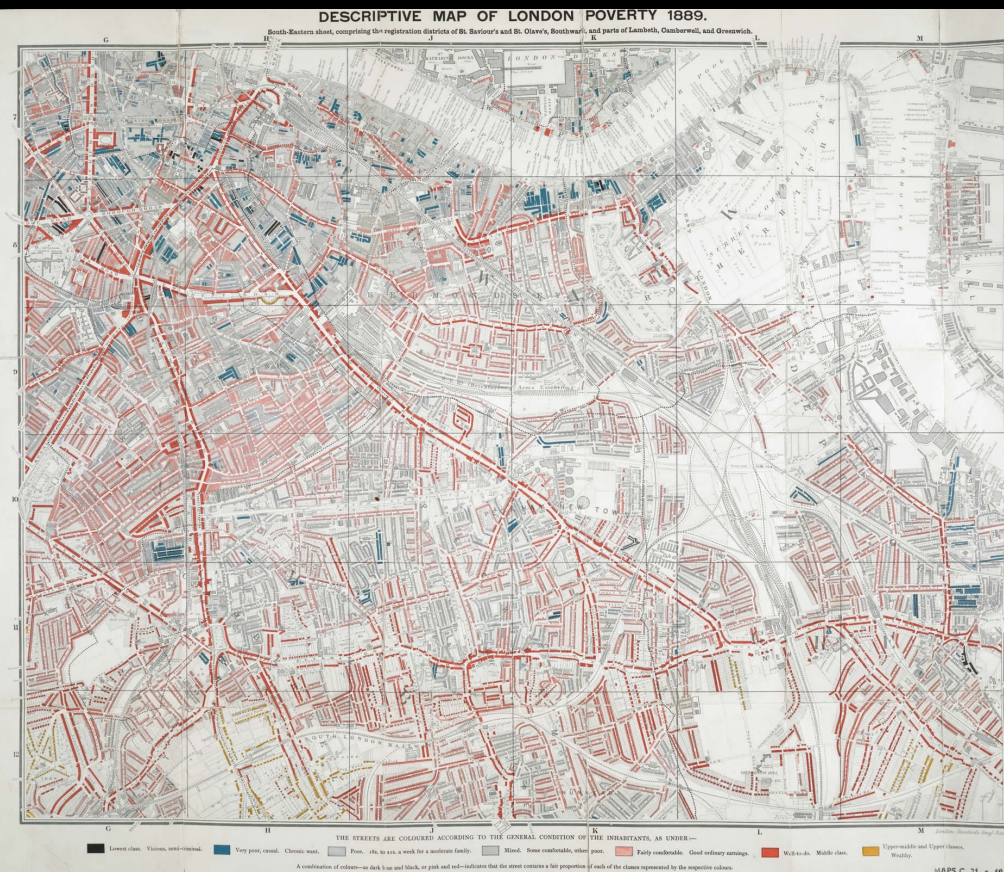
Cholera map in London, 1854, John Snow



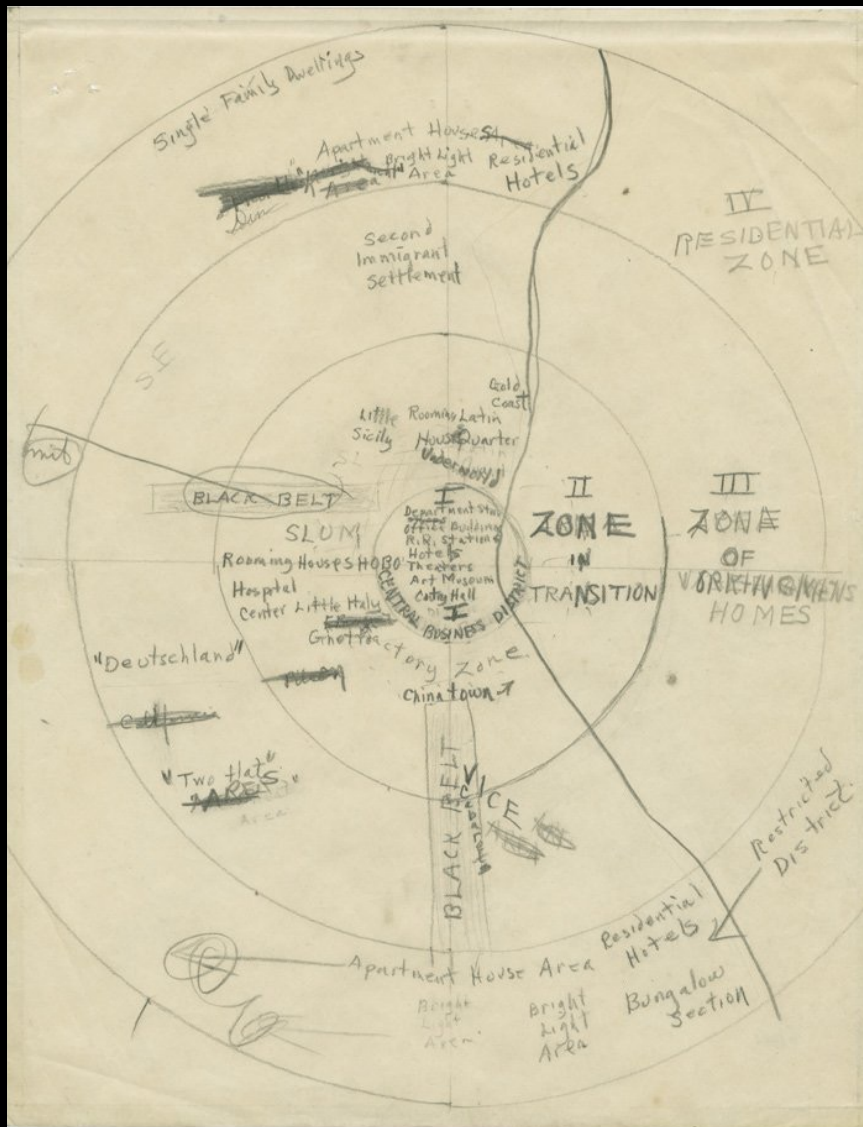
John Snow



Is urban poverty caused by individuals?



Descriptive Map of London Poverty, 1889, Charles Booth

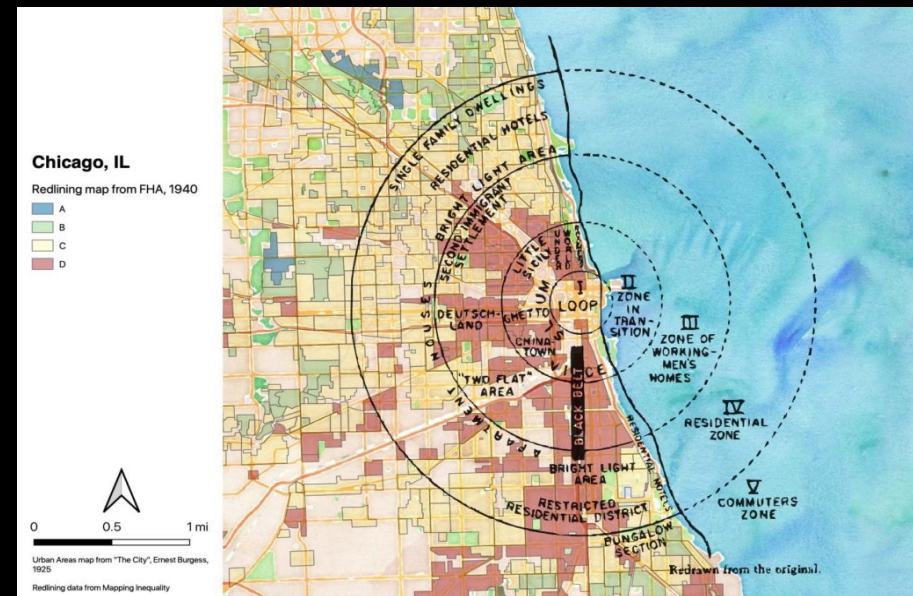


Map of the Radial Expansion and the Five Urban Zones, early 20th Century, Ernest Burgess

How does a city form?

"In every course I gave I am sure there were one or two students who made maps. I think the maps of juvenile delinquency were the first ones undertaken. They were followed by maps showing the distribution of motion picture houses. Then came maps showing the distribution of patrons of the public dance halls. . . We were very impressed with the great differences between the various neighborhoods in the city, and one of the earliest goals was to try to find a pattern to this patchwork of differences, and to 'make sense of it.'"

Ernest Burgess, 1964



<https://bike-lab.org/2020/08/12/burgess-concentric-circle-map-in-gis/>

A TAXONOMY OF TRANSITIONS

racial / ethnic
self-identification
in chicago
in the year 2000

white ■
black ■
asian ■
hispanic ■
other ■

the black lines show
chicago's official
community areas.

each dot represents
twenty-five people.
here, hispanic is
exclusive of other
categories.

block-level data
from the U.S. census.

scale 1:200,000

The same data, aggregated by community
area and shown with solid colors.

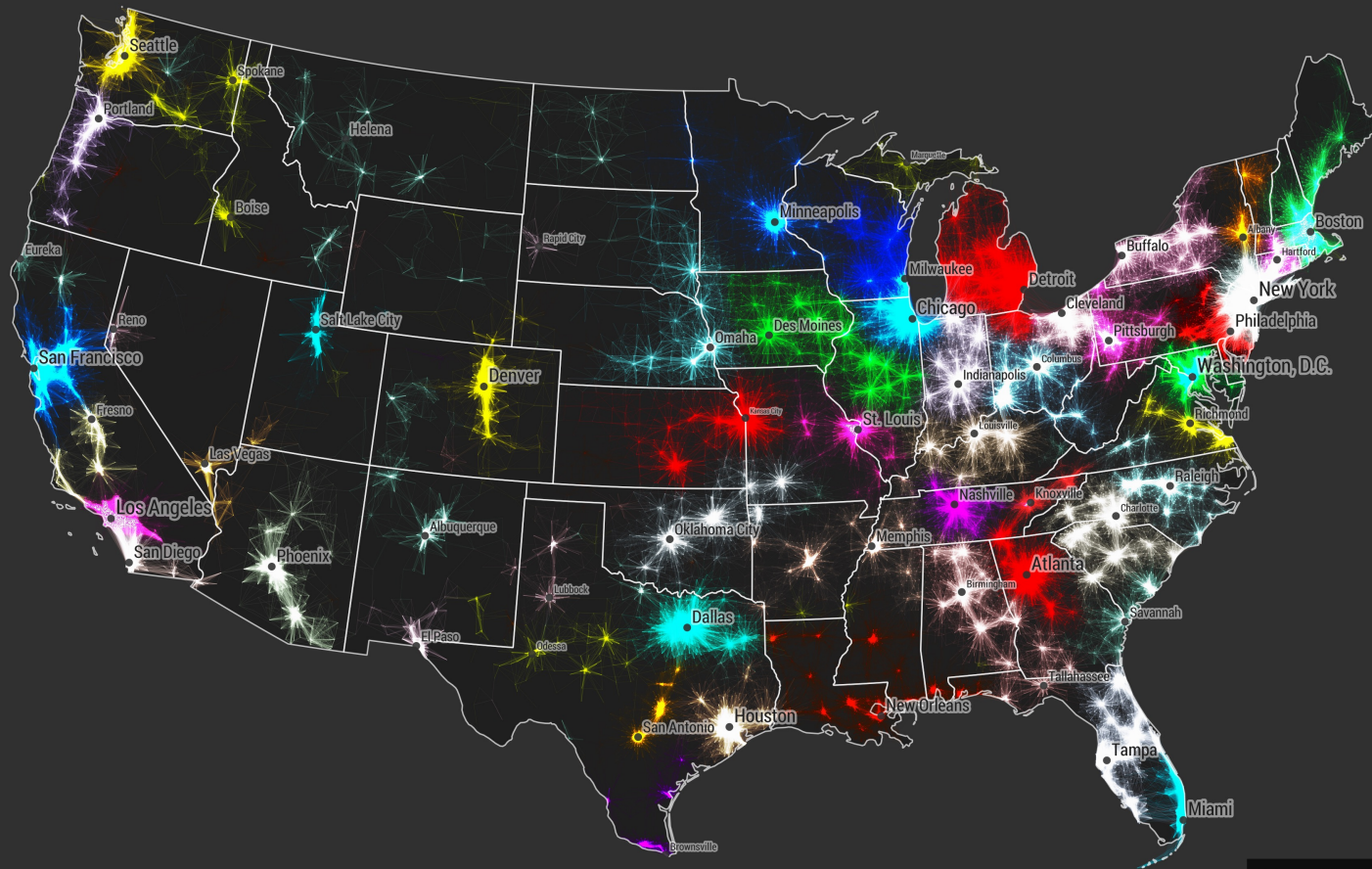
■ >80% white
■ majority white
■ >80% hispanic
■ majority hispanic
■ >80% black
■ majority black
■ majority asian
■ majority asian
■ no majority

To what extent does residential segregation exist?

his map reveals the racial makeup of cities across the country, visualizing trends such as segregation and density. A researcher with the the University of Virginia's Weldon Cooper Center for Public Service, modeled the project on a previous MIT map plotting population density by individual dots. This version color-codes the results by race and ethnicity, producing an eerily beautiful picture of American segregation (and, less frequently, integration) that tricks the eye at different scales. At most zoom levels, each dot is smaller than a pixel, and so the blended colors from afar are "aggregations of many individual dots," with people represented by the color scheme at right. Looking at the entire country, most of the patches that aren't blue correspond to colorfully smudged urban areas.

<http://www.radicalcartography.net/index.html?chicagodots>

Commuter Megaregions of the United States



Where should planners draw the edges of a megaregion encompassing this activity?

Which connections are statistically significant?

Which are important for regional transit planning?

This study uses a data set of more than 4,000,000 commuter flows as the basis for an empirical approach to the identification of such megaregions.

This made for a nice test of Waldo Tobler's "first law of geography": that things that are near each other are more related than those that are farther apart.

Explore Shade Distribution

shade and income levels in each neighborhood across the city

Urban shade is a critical resource.
But is it equally accessible across cities?



Select a city

Hong Kong



*Income level is based on city specific metrics

<https://senseable.mit.edu/shaded-politics/>

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